

Agenda LSTM on paper

Inshorts

1000s $\rightarrow$ 60-70

"I grew up in France, so I speak fluent —"

$$\underline{RNN} \rightarrow \textcircled{h_t} = \tanh(W_h h_{t-1} + W_x x_t + b_h)$$

word 1
↓

"I grew up in France, and later I moved to Canada, learned English, and now I speak fluent —"

word 50
↓

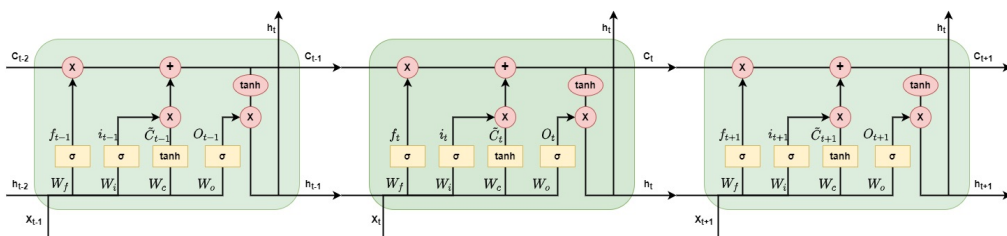
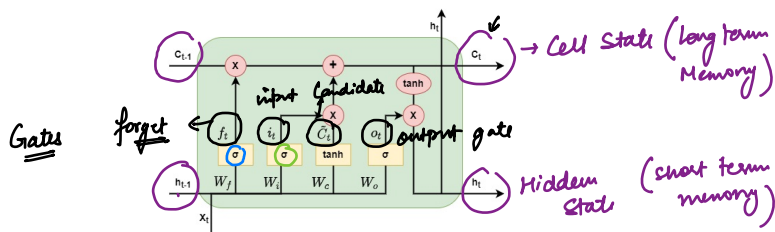
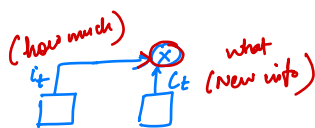
$$\frac{\partial L}{\partial h_{T-1}} = \frac{\partial L}{\partial h_T} \times \frac{\partial h_T}{\partial h_{T-1}} \dots \dots \dots$$

BPTT

Vanishing / exploding gradient

LSTM

Long - Short Term Memory



France

Initializing

$$h_{T-1} = \underline{0.5} \quad (\text{chart})$$

long term memory

$$C_{t-1} = \underline{0.8}$$

$$x_t = 1.2$$

Gates

Forget Gate $\rightarrow$ "How much of France info to keep?"

$$\begin{aligned}
 z &= w_f^{(h)} h_{t-1} + w_f^{(x)} x_t + b_f \\
 &= 0.7 \times 0.5 + 0.4 \times 1.2 + 0.5
 \end{aligned}$$

$$= 1.33$$

$$f_t = \sigma(1.33) \Rightarrow \underline{\underline{0.8}}$$

Keep 80% of the frame info
old cell state
info

If $f_t = 0 \rightarrow$ forget frame info

Candidate Memory $\bar{C}_t \rightarrow$ "what new info we could add?"

$$\chi_t = \underbrace{W_c^{(1)}}_{-0.6} h_{t-1} + \underbrace{W_c^{(2)}}_{0.1} x_t + \underbrace{b_c}_{1.0}$$

$$\Rightarrow -0.6 \times 0.5 + 0.1 \times 1.2 + 0$$

$$\Rightarrow -0.18$$

$$\bar{C}_t = \tanh(-0.18) = \underline{\underline{-0.18}}$$

Input gate

$$\chi_i = \underbrace{W_i^h}_{0.2} h_{t-1} + \underbrace{W_i^x}_{0.5} x_t + \underbrace{b_i}_{-0.1}$$

$$\chi_i = 0.6$$

$$\sigma(\chi_i) = \sigma(0.6) = \underline{\underline{0.65}}$$

Update the Cell state

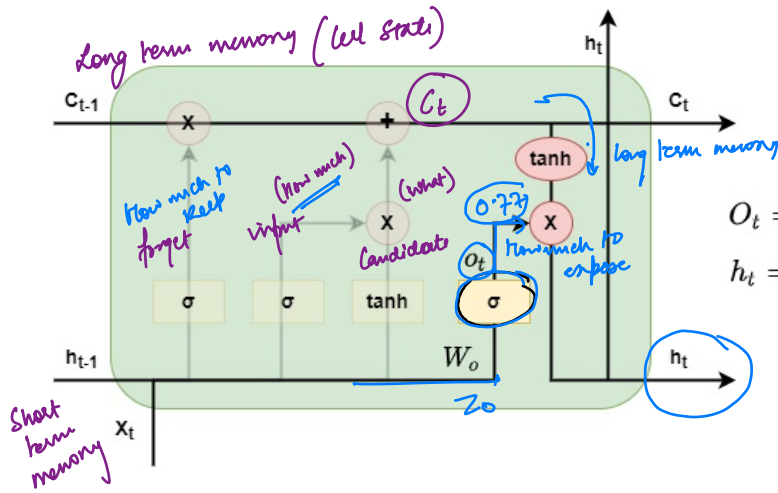
$$C_t = \underbrace{f_t}_{\text{how much to add}} \times \underbrace{C_{t-1}}_{\text{new info}} + \underbrace{i_t \times \bar{C}_t}_{\text{new info}}$$

0.8
 Class 11
 Class 12
 IIT Exam

$$C_t = 80\% [0.8] + 65\% [-0.18]$$

$$= 0.8 \times 0.8 + 0.65 \times (-0.18)$$

$$= 0.515$$



$$O_t = \sigma(W_o * [h_{t-1}, x_t] + b_o)$$

$$h_t = O_t * \tanh(C_t)$$

Output gate → How much of C_t to show?

Model has to decide that how much of long term memory (C_t) to expose to short term memory (h_t) for prediction of next word.

$$Z_o = W_o^{(h)} h_{t-1} + W_o^{(x)} x_t + b_o$$

$$= 0.3 \times 0.5 + 0.7 \times 1.7 + 0.2$$

$$= 1.19$$

$$\underline{O_t} = \sigma(1.19) \approx 0.77$$

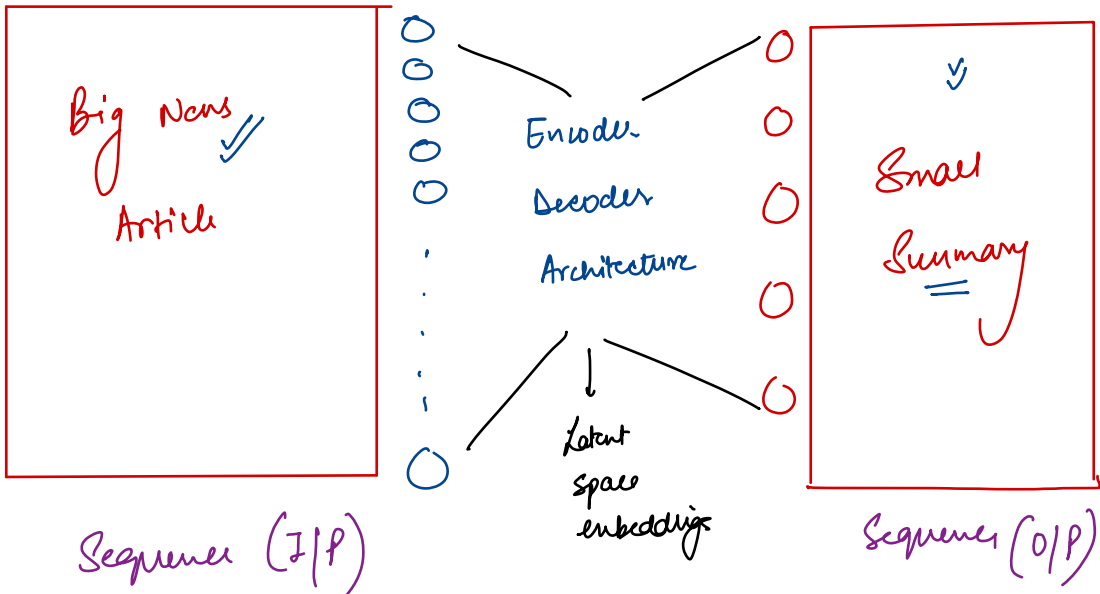
" 77% of the info should be exposed "

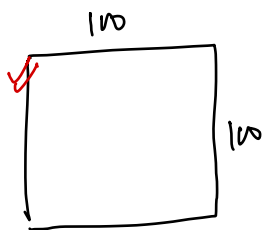
$$\textcircled{H_t} = O_t \times \underbrace{\tanh(C_t)}_{\text{long term memory}}$$

$$\Rightarrow \underline{0.77} \times \tanh(\underline{0.515})$$

$$= 0.77 \times \underline{0.47} \approx \underline{\underline{0.36}}$$

$$h_t = \underline{\underline{0.36}}$$





$x_1, 0$
 0
 0
 0
 0
 $\vdots$
 $\vdots$
 $\vdots$
 $\vdots$
 $x_{10,000}$



⊕

0

0

0

0

Some info that represent info in 10,000 i/p neurons

Latent Space embeddings Summary (10,000)

0

0

0



```

# ----- Encoder -----
encoder_inputs = Input(shape=(max_text_len,))
enc_emb = Embedding(x_vocab_size, embedding_dim, trainable=True)(encoder_inputs)
encoder_lstm = LSTM(latent_dim, return_state=True)
encoder_outputs, state_h, state_c = encoder_lstm(enc_emb)
# state_h, state_c are h_T^enc, C_T^enc

# ----- Decoder -----
decoder_inputs = Input(shape=(None,))
dec_emb_layer = Embedding(y_vocab_size, embedding_dim, trainable=True)
dec_emb = dec_emb_layer(decoder_inputs)

decoder_lstm = LSTM(latent_dim, return_sequences=True, return_state=True)
decoder_outputs, _, _ = decoder_lstm(dec_emb, initial_state=[state_h, state_c])

```

Big Articles

Small Articles

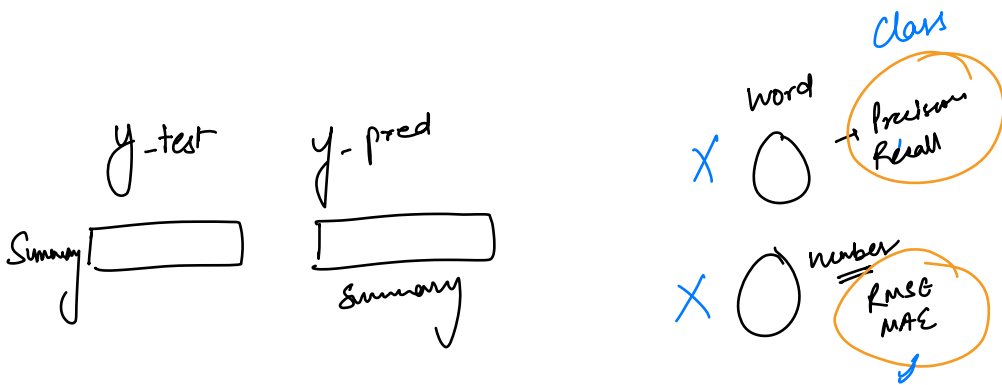
Encoder → Reader → Reads full Articles → stores the notes in internal memory

Ct

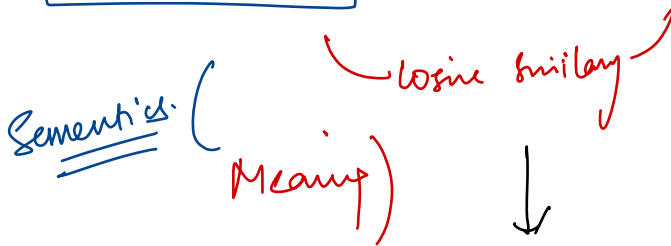
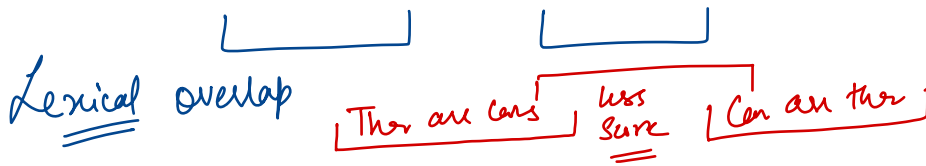
Decoder → Writer → Notes → Small Summary

Hr

Lower word at a time



Rouge



BERTSCORE

RNN $\rightarrow$ LSTM $\rightarrow$ GRU